

Malware Analysis Report For libSystem.B

Binary Type

Mach-O (Mach Object). (Approved)

Analyzing File

- **File Name:** libSystem.B
 - **File Extension:** .dylib
 - **File Type:** Mach-O (Mach Object).
 - **MD5 Hash:** 2fc99dfb535b42db2410f72de63df31c
 - **SHA256 Hash:** 0e11dcdc203cee5bb59652bd971305fe8893559779ec1d6ad3f9b3234efa55d0
-

File Identification

File detected as a valid macOS/iOS executable.

Mach-O File Analysis

Mach-O File Analysis: True

Mach-O File Imports

No imported functions were found.

Mach-O File Exports

No exported functions were found.

String Analysis

ASCII Strings

- __TEXT
- __text
- __TEXT
- __stubs
- __TEXT
- __stubhelper
- __TEXT
- __const
- __TEXT
- __unwindinfo
- __TEXT
- __DATA
- _nlsymbol_ptr
- __DATA

- `_/_asymbol_ptr`
- `__DATA`
- `_modinit_func`
- `__DATA`
- `__const`
- `__DATA`
- `__data`
- `__DATA`
- `__common`
- `__DATA`
- `__LINKEDIT`
- `/usr/lib/libSystem.B.dylib`
- `/usr/lib/system/libcache.dylib`
- `/usr/lib/system/libcommonCrypto.dylib`
- `/usr/lib/system/libcompiler_rt.dylib`
- `/usr/lib/system/libcopyfile.dylib`
- `/usr/lib/system/libcorecrypto.dylib`
- `/usr/lib/system/libdispatch.dylib`
- `/usr/lib/system/libdyld.dylib`
- `/usr/lib/system/libkeymgr.dylib`
- `/usr/lib/system/liblaunch.dylib`
- `/usr/lib/system/libmacho.dylib`
- `/usr/lib/system/libquarantine.dylib`
- `/usr/lib/system/libremovefile.dylib`
- `/usr/lib/system/libsystem_asl.dylib`
- `/usr/lib/system/libsystem_blocks.dylib`
- `/usr/lib/system/libsystem_c.dylib`
- `/usr/lib/system/libsystem_configuration.dylib`
- `/usr/lib/system/libsystem_dnssd.dylib`
- `/usr/lib/system/libsystem_info.dylib`
- `/usr/lib/system/libsystem_kernel.dylib`
- `/usr/lib/system/libsystem_m.dylib`
- `/usr/lib/system/libsystem_malloc.dylib`
- `/usr/lib/system/libsystem_network.dylib`
- `/usr/lib/system/libsystem_notify.dylib`
- `/usr/lib/system/libsystem_platform.dylib`
- `/usr/lib/system/libsystem_pthread.dylib`
- `/usr/lib/system/libsystem_sandbox.dylib`
- `/usr/lib/system/libsystem_stats.dylib`
- `/usr/lib/system/libunc.dylib`
- `/usr/lib/system/libunwind.dylib`
- `/usr/lib/system/libxpc.dylib`
- `AWAVSPM`
- `[A^A_]`
- `@(#)PROGRAM:System.B PROJECT:Libsystem-1197.1.1`
- `0@_pthreadexitifcanceled`
- `@_dlsym`
- `@_exit`
- `@_free`
- `@_malloc`
- `@_realloc`
- `@dyldstubbinder`

- 0@dispatchfork_child
- 0@dispatchfork_parent
- q 0@dispatchfork_prepare
- q(0@libdispatchinit
- q00@_dyldfork_child
- q80@_dyldinitializer
- q@0@_keymgr_initializer
- qH0@bootstrapinit
- qP0@_asfork_child
- qX0@_libc_init
- q`0@_libcfork_child
- qh0@_fork
- qp0@_sigaction
- qx0@_sysctl
- 0@_time
- 0@_libSCinfoforkchild
- 0@_libSCinfoforkparent
- 0@_libSCinfoforkprepare
- 0@_error
- 0@_libkernel_init
- 0@_machfork_child
- 0@_close
- 0@_fsync
- 0@_getattrlist
- 0@_getrlimit
- 0@_getxattr
- 0@_open
- 0@_ptrace
- 0@_read
- 0@_setattrlist
- 0@_setrlimit
- 0@_stat
- 0@_unlink
- 0@_write
- 0@_malloc_init
- 0@_stackloggingearly_finished
- 0@_mallocfork_child
- 0@_mallocfork_parent
- 0@_mallocfork_prepare
- 0@_notifyfork_child
- 0@_libplatform_init
- 0@_pthread_init
- 0@_pthreadfork_child
- 0@_pthreadforkchildpostinit
- 0@_pthreadfork_parent
- 0@_pthreadfork_prepare
- 0@pthreadattr_destroy
- 0@pthreadattr_init
- 0@pthreadattr_setdetachstate
- 0@pthreadcreate
- 0@pthreadmutex_lock

- 0@pthreadmutex_unlock
- 0@pthreadself
- 0@_libxpcinitializer
- 0@xpcatfork_child
- 0@xpcatfork_parent
- 0@xpcatfork_prepare
- R8289209\$_
- unlink
- attrlist
- rlimit
- hread_
- create
- mutex_
- destroy
- setdetachstate
- unlock
- igaction
- attrlist
- rlimit
- libSystematfork
- machinitroutine
- repare
- System_BVersion
- crashreporterinfo_
- String
- Number
- com.apple.libcache
- com.apple.libcommonCrypto
- com.apple.libcompiler_rt
- com.apple.libcopyfile
- com.apple.libcorecrypto
- com.apple.libdispatch
- com.apple.libdyld
- com.apple.libkeymgr
- com.apple.liblaunch
- com.apple.libmacho
- com.apple.libquarantine
- com.apple.libremovefile
- com.apple.libsystem_asl
- com.apple.libsystem_blocks
- com.apple.libsystem_c
- !com.apple.libsystem_configuration
- com.apple.libsystem_dnssd
- com.apple.libsystem_info
- com.apple.libsystem_kernel
- com.apple.libsystem_m
- com.apple.libsystem_malloc
- com.apple.libsystem_network
- com.apple.libsystem_notify
- com.apple.libsystem_platform
- com.apple.libsystem_pthread
- com.apple.libsystem_sandbox

- com.apple.libsystem_stats
- com.apple.libunc
- com.apple.libunwind
- com.apple.libxpc
- R8289209\$_close
- R8289209\$_fork
- R8289209\$_fsync
- R8289209\$_getattrlist
- R8289209\$_getrlimit
- R8289209\$_getxattr
- R8289209\$_open
- R8289209\$pthreadattr_destroy
- R8289209\$pthreadattr_init
- R8289209\$pthreadattr_setdetachstate
- R8289209\$pthreadcreate
- R8289209\$pthreadmutex_lock
- R8289209\$pthreadmutex_unlock
- R8289209\$pthreadself
- R8289209\$_ptrace
- R8289209\$_read
- R8289209\$_setattrlist
- R8289209\$_setrlimit
- R8289209\$_sigaction
- R8289209\$_stat
- R8289209\$_sysctl
- R8289209\$_time
- R8289209\$_unlink
- R8289209\$_write
- _System_BVersionNumber
- _System_BVersionString
- **crashreporterinfo**
- libSystematfork_child
- libSystematfork_parent
- libSystematfork_prepare
- machinit_routine
- _error
- _keymgr_initializer
- _libc_init
- _libkernel_init
- _libplatform_init
- _malloc_init
- _pthread_init
- _stackloggingearly_finished
- _as/fork_child
- _dyldfork_child
- _dyldinitializer
- _libSCinfoforkchild
- _libSCinfoforkparent
- _libSCinfoforkprepare
- _libcfork_child
- _libxpcinitializer

- *_machfork_child*
- *_mallocfork_child*
- *_mallocfork_parent*
- *_mallocfork_prepare*
- *_notifyfork_child*
- *_pthreadexitifcanceled*
- *_pthreadfork_child*
- *_pthreadforkchildpostinit*
- *_pthreadfork_parent*
- *_pthreadfork_prepare*
- *bootstrapinit*
- *_close*
- *dispatchatfork_child*
- *dispatchatfork_parent*
- *dispatchatfork_prepare*
- *_dlsym*
- *_fsync*
- *_getattrlist*
- *_getrlimit*
- *_getxattr*
- *libdispatchinit*
- *_malloc*
- *pthreadattr_destroy*
- *pthreadattr_init*
- *pthreadattr_setdetachstate*
- *pthreadcreate*
- *pthreadmutex_lock*
- *pthreadmutex_unlock*
- *pthreadself*
- *_ptrace*
- *_realloc*
- *_setattrlist*
- *_setrlimit*
- *_sigaction*
- *_sysctl*
- *_unlink*
- *_write*
- *xpcatfork_child*
- *xpcatfork_parent*
- *xpcatfork_prepare*
- *dyldstubfinder*
- *libSysteminitializer*
- *machinit_old*
- *libSysteminitializer.libkernel_funcs*
- *libSysteminitializer.libpthread_funcs*
- *libSystem.B*
- *libSystem.B*
- *com.apple.libcache*
- *com.apple.libcommonCrypto*
- *com.apple.libcompiler_rt*
- *com.apple.libcopyfile*

- com.apple.libcorecrypto
- com.apple.libdispatch
- com.apple.libdyld
- com.apple.libkeymgr
- com.apple.liblaunch
- com.apple.libmacho
- com.apple.libquarantine
- com.apple.libremovefile
- com.apple.libsystem_asl
- com.apple.libsystem_blocks
- com.apple.libsystem_c
- !com.apple.libsystem_configuration
- com.apple.libsystem_dnssd
- com.apple.libsystem_info
- com.apple.libsystem_kernel
- com.apple.libsystem_m
- com.apple.libsystem_malloc
- com.apple.libsystem_network
- com.apple.libsystem_notify
- com.apple.libsystem_platform
- com.apple.libsystem_pthread
- com.apple.libsystem_sandbox
- com.apple.libsystem_stats
- com.apple.libunc
- com.apple.libunwind
- com.apple.libxpc
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- 111024173941Z
- 261024173941Z0
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- drmljGa
- %http://www.apple.com/appleca/root.crl0
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- 060425214036Z
- 350209214036Z0b1
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- https://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- 130412223435Z
- 210412223435Z0V1
- Apple Inc.1

- Apple Software1
- Software Signing0
- Ztd9?Yn
-)http://www.apple.com/certificateauthority0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- \$http://crl.apple.com/codesigning.crl0
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority
- 20(~ht
- __TEXT
- __text
- __TEXT
- _symbolstub
- __TEXT
- _stubhelper
- __TEXT
- __const
- __TEXT
- _unwindinfo
- __TEXT
- __DATA
- _n/symbol_ptr
- __DATA
- _/asymbol_ptr
- __DATA
- _modinit_func
- __DATA
- __const
- __DATA
- __data
- __DATA
- __common
- __DATA
- __LINKEDIT
- /usr/lib/libSystem.B.dylib
- /usr/lib/system/libcache.dylib
- /usr/lib/system/libcommonCrypto.dylib
- /usr/lib/system/libcompiler_rt.dylib
- /usr/lib/system/libcopyfile.dylib
- /usr/lib/system/libcorecrypto.dylib
- /usr/lib/system/libdispatch.dylib
- /usr/lib/system/libdyld.dylib
- /usr/lib/system/libkeymgr.dylib
- /usr/lib/system/liblaunch.dylib
- /usr/lib/system/libmacho.dylib
- /usr/lib/system/libquarantine.dylib
- /usr/lib/system/libremovefile.dylib
- /usr/lib/system/libsystem_asl.dylib
- /usr/lib/system/libsystem_blocks.dylib
- /usr/lib/system/libsystem_c.dylib
- /usr/lib/system/libsystem_configuration.dylib

- /usr/lib/system/libsystem_dnssd.dylib
- /usr/lib/system/libsystem_info.dylib
- /usr/lib/system/libsystem_kernel.dylib
- /usr/lib/system/libsystem_m.dylib
- /usr/lib/system/libsystem_malloc.dylib
- /usr/lib/system/libsystem_network.dylib
- /usr/lib/system/libsystem_notify.dylib
- /usr/lib/system/libsystem_platform.dylib
- /usr/lib/system/libsystem_pthread.dylib
- /usr/lib/system/libsystem_sandbox.dylib
- /usr/lib/system/libunc.dylib
- /usr/lib/system/libunwind.dylib
- /usr/lib/system/libxpc.dylib
- @(#)PROGRAM:System.B PROJECT:Libsystem-1197.1.1
- 0@_pthreadexitifcanceled
- @_dlsym
- @_exit
- @_free
- @_malloc
- @_realloc
- @dyldstubfinder
- 0@dispatchfork_child
- 0@dispatchfork_parent
- 0@dispatchfork_prepare
- 0@libdispatchinit
- 0@_dyldfork_child
- 0@_dyldinitializer
- q 0@_keymgr_initializer
- q\$0@bootstrapinit
- q(0@_asfork_child
- q,0@_libc_init
- q00@_libcfork_child
- q40@_fork
- q80@_sigaction
- q<0@_sysctl
- q@0@_time
- qD0@_libSCinfoforkchild
- qH0@_libSCinfoforkparent
- qL0@_libSCinfoforkprepare
- qP0@_error
- qT0@_libkernel_init
- qX0@_machfork_child
- q\0@_close
- q`0@_fsync
- qd0@_getattrlist
- qh0@_getrlimit
- ql0@_getxattr
- qp0@_open
- qt0@_ptrace
- qx0@_read
- q|0@_setattrlist

- 0@_setrlimit
- 0@_stat
- 0@_unlink
- 0@_write
- 0@_malloc_init
- 0@_stackloggingearly_finished
- 0@_mallocfork_child
- 0@_mallocfork_parent
- 0@_mallocfork_prepare
- 0@_notifyfork_child
- 0@_libplatform_init
- 0@_pthread_init
- 0@_pthreadfork_child
- 0@_pthreadforkchildpostinit
- 0@_pthreadfork_parent
- 0@_pthreadfork_prepare
- 0@pthreadattr_destroy
- 0@pthreadattr_init
- 0@pthreadattr_setdetachstate
- 0@pthreadcreate
- 0@pthreadmutex_lock
- 0@pthreadmutex_unlock
- 0@pthreadself
- 0@_libxpcinitializer
- 0@xpcatfork_child
- 0@xpcatfork_parent
- 0@xpcatfork_prepare
- R8289209\$_
- unlink
- attrlist
- rlimit
- hread_
- create
- mutex_
- destroy
- setdetachstate
- unlock
- igaction
- attrlist
- rlimit
- libSystematfork
- machinitroutine
- repare
- System_BVersion
- crashreporterinfo_
- String
- Number
- com.apple.libcache
- com.apple.libcommonCrypto
- com.apple.libcompiler_rt
- com.apple.libcopyfile

- com.apple.libcorecrypto
- com.apple.libdispatch
- com.apple.libdyld
- com.apple.libkeymgr
- com.apple.liblaunch
- com.apple.libmacho
- com.apple.libquarantine
- com.apple.libremovefile
- com.apple.libsystem_asl
- com.apple.libsystem_blocks
- com.apple.libsystem_c
- !com.apple.libsystem_configuration
- com.apple.libsystem_dnssd
- com.apple.libsystem_info
- com.apple.libsystem_kernel
- com.apple.libsystem_m
- com.apple.libsystem_malloc
- com.apple.libsystem_network
- com.apple.libsystem_notify
- com.apple.libsystem_platform
- com.apple.libsystem_pthread
- com.apple.libsystem_sandbox
- com.apple.libunc
- com.apple.libunwind
- com.apple.libxpc
- R8289209\$_close
- R8289209\$_fork
- R8289209\$_fsync
- R8289209\$_getattrlist
- R8289209\$_getrlimit
- R8289209\$_getxattr
- R8289209\$_open
- R8289209\$pthreadattr_destroy
- R8289209\$pthreadattr_init
- R8289209\$pthreadattr_setdetachstate
- R8289209\$pthreadcreate
- R8289209\$pthreadmutex_lock
- R8289209\$pthreadmutex_unlock
- R8289209\$pthreadself
- R8289209\$_ptrace
- R8289209\$_read
- R8289209\$_setattrlist
- R8289209\$_setrlimit
- R8289209\$_sigaction
- R8289209\$_stat
- R8289209\$_sysctl
- R8289209\$_time
- R8289209\$_unlink
- R8289209\$_write
- _System_BVersionNumber
- _System_BVersionString
- **crashreporterinfo**

- *libSystematfork_child*
- *libSystematfork_parent*
- *libSystematfork_prepare*
- *machinit_routine*
- *_error*
- *_keymgr_initializer*
- *_libc_init*
- *_libkernel_init*
- *_libplatform_init*
- *_malloc_init*
- *_pthread_init*
- *_stackloggingearly_finished*
- *_as/fork_child*
- *_dyldfork_child*
- *_dyldinitializer*
- *_libSCinfoforkchild*
- *_libSCinfoforkparent*
- *_libSCinfoforkprepare*
- *_libcfork_child*
- *_libxpcinitializer*
- *_machfork_child*
- *_mallocfork_child*
- *_mallocfork_parent*
- *_mallocfork_prepare*
- *_notifyfork_child*
- *_pthreadexitifcanceled*
- *_pthreadfork_child*
- *_pthreadforkchildpostinit*
- *_pthreadfork_parent*
- *_pthreadfork_prepare*
- *bootstrapinit*
- *_close*
- *dispatchatfork_child*
- *dispatchatfork_parent*
- *dispatchatfork_prepare*
- *_dlsym*
- *_fsync*
- *_getattrlist*
- *_getrlimit*
- *_getxattr*
- *libdispatchinit*
- *_malloc*
- *pthreadattr_destroy*
- *pthreadattr_init*
- *pthreadattr_setdetachstate*
- *pthreadcreate*
- *pthreadmutex_lock*
- *pthreadmutex_unlock*
- *pthreadself*
- *_ptrace*

- `_realloc`
- `_setattrlist`
- `_setrlimit`
- `_sigaction`
- `_sysctl`
- `_unlink`
- `_write`
- `xpcatfork_child`
- `xpcatfork_parent`
- `xpcatfork_prepare`
- `dyldstubbinder`
- `libSysteminitializer`
- `machinit_old`
- `libSysteminitializer.libkernel_funcs`
- `libSysteminitializer.libpthread_funcs`
- `libSystem.B`
- `libSystem.B`
- `com.apple.libcache`
- `com.apple.libcommonCrypto`
- `com.apple.libcompiler_rt`
- `com.apple.libcopyfile`
- `com.apple.libcorecrypto`
- `com.apple.libdispatch`
- `com.apple.libdyld`
- `com.apple.libkeymgr`
- `com.apple.liblaunch`
- `com.apple.libmacho`
- `com.apple.libquarantine`
- `com.apple.libremovefile`
- `com.apple.libsystem_asl`
- `com.apple.libsystem_blocks`
- `com.apple.libsystem_c`
- `!com.apple.libsystem_configuration`
- `com.apple.libsystem_dnssd`
- `com.apple.libsystem_info`
- `com.apple.libsystem_kernel`
- `com.apple.libsystem_m`
- `com.apple.libsystem_malloc`
- `com.apple.libsystem_network`
- `com.apple.libsystem_notify`
- `com.apple.libsystem_platform`
- `com.apple.libsystem_pthread`
- `com.apple.libsystem_sandbox`
- `com.apple.libsystem_stats`
- `com.apple.libunc`
- `com.apple.libunwind`
- `com.apple.libxpc`
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- 111024173941Z
- 261024173941Z0

- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- drmljGa
- %http://www.apple.com/appleca/root.crl0
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- 060425214036Z
- 350209214036Z0b1
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- https://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- 130412223435Z
- 210412223435Z0V1
- Apple Inc.1
- Apple Software1
- Software Signing0
- Ztd9?Yn
-)http://www.apple.com/certificateauthority0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- \$http://crl.apple.com/codesigning.crl0
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority

UTF-8 Strings

No detected.

UTF-16LE Strings

No detected.

Extracted URLs

- http://www.apple.com/appleca/root.crl0
- https://www.apple.com/appleca/0
- http://www.apple.com/certificateauthority0
- http://crl.apple.com/codesigning.crl0
- http://www.apple.com/appleca/root.crl0
- https://www.apple.com/appleca/0
- http://www.apple.com/certificateauthority0
- http://crl.apple.com/codesigning.crl0

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy:** 3.99
- **Status:** This file likely contains Standard Text or human-readable text.

VirusTotal Analysis

Scan Date: 2021-08-11 14:30:01

Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 59
- Harmless: 0
- Timeout: 2
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 13

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

Engine Name	Detection Category	Detection
No results	No results	No results